# IGCSE 0607 Practice Questions — E10.6 — Combined Questions & Markscheme

## Statistical charts and diagrams

**Total Questions:** 9 — **Total Marks:** 60 Marks

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**1** The stem-and-leaf diagram shows the masses, in kg, of 15 parcels.

| | | | | | |
|---|---|---|---|---|---|
| 1 | 2 | 4 | 7 | | |
| 2 | 0 | 1 | 1 | 1 | 5 |
| 3 | 0 | 2 | 6 | 8 | |
| 4 | 1 | 4 | 5 | | |

**Key:** 1 | 2 represents 12 kg

**(a)** Find the mode of the masses.

..... kg **[1]**

**(b)** Find the median mass.

..... kg **[1]**

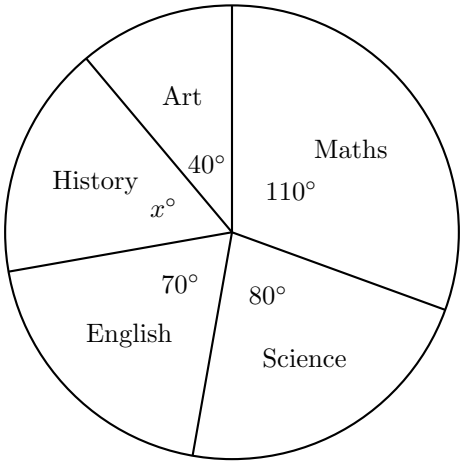
**(c)** Work out the range of the masses.

..... kg **[1]**

**(d)** A parcel is chosen at random.
Find the probability that its mass is strictly greater than 30 kg.

..... **[2]**

**2** The pie chart shows information about the favourite subjects of 180 students.




**(a)** Find the value of $x$.

$x =$ ..... **[1]**

**(b)** Find the number of students whose favourite subject is Science.

..... **[2]**

**(c)** Write down the fraction of students whose favourite subject is Maths.
Give your fraction in its simplest form.

..... **[2]**

**3** The masses, $m$ kilograms, of 15 parcels are recorded.

3.2 1.5 4.6 2.1 1.8 3.5 2.9 4.1 3.2 1.9 2.4 3.7 4.0 2.6 1.7

**(a)** Complete the ordered stem-and-leaf diagram to show this information.

| | |
|---|---|
| 1 | |
| 2 | |
| 3 | |
| 4 | |

Key: 1 | 5 represents 1.5 kg

**[3]**

**(b)** Find the median mass.

..... kg **[1]**

**(c)** A parcel is chosen at random.
Find the probability that its mass is greater than 3.5 kg.

..... **[1]**

**4** A survey asked 120 students about their favourite sport.
The results are shown in the table.

| **Sport** | **Frequency** |
|---|---|
| Football | 45 |
| Basketball | 35 |
| Tennis | 25 |
| Swimming | 15 |

**(a)** A pie chart is drawn to show this information.
Calculate the sector angle for Basketball.

..... ° **[2]**

**(b)** The pie chart has a radius of 6 cm.

Calculate the area of the sector representing Tennis.

..... $cm^2$ **[3]**

**(c)** In a different school, the pie chart for favourite sports has a sector angle of 108° for Football.
This sector represents 54 students.
Calculate the total number of students in this school.

..... **[2]**

**5** A pie chart is drawn to show the favourite sport of a group of students.
The angles of the sectors for each sport are given in terms of $x$.

Football: $(2x + 30)^\circ$ Basketball: $x^\circ$ Tennis: $(x + 50)^\circ$ Swimming: $(2x - 20)^\circ$

**(a)** Find the value of $x$.

$x =$ ..... **[2]**

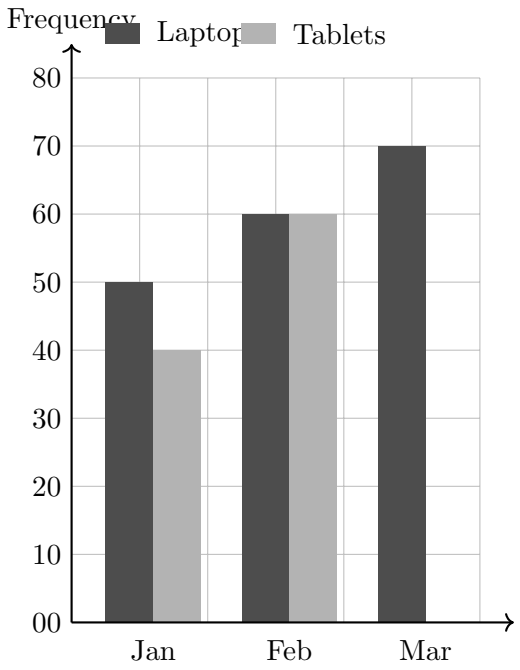
**(b)** 10 students chose Tennis as their favourite sport.
Calculate the total number of students in the group.

..... **[2]**

**(c)** Two students are chosen at random from the group.
Calculate the probability that one student chose Football and the other student chose Basketball.

..... **[3]**

**6** A company records the number of Laptops and Tablets sold over three months.
The dual bar chart shows the sales for January, February, and March.
The bar for Tablets sold in March is incomplete and not shown.




**(a)** The total number of Laptops sold over the three months is 20% more than the total number of Tablets sold.
Calculate the number of Tablets sold in March.

..... **[4]**

**(b)** In April, the total number of items sold is $\frac{1}{3}$ of the total number of items sold in January, February, and March combined.
The ratio of Laptops to Tablets sold in April is 6 : 5.
Work out the number of Laptops sold in April.

..... **[4]**

**(c)** The price of a Laptop is \$450 and the price of a Tablet is \$200.
Calculate the percentage of the total revenue in January that came from Tablet sales.

..... % **[3]**

**7** The masses, in kilograms, of 15 parcels are recorded below.

| | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 27 | 15 | 31 | 42 | 23 | 12 | 53 | 38 | 49 | 27 | 45 | 40 | 34 | 51 | 42 |

**(a)** Complete the ordered stem-and-leaf diagram to show this information.

| | |
|---|---|
| 1 | |
| 2 | |
| 3 | |
| 4 | |
| 5 | |

Key: 1 | 2 represents 12 kg

**[3]**

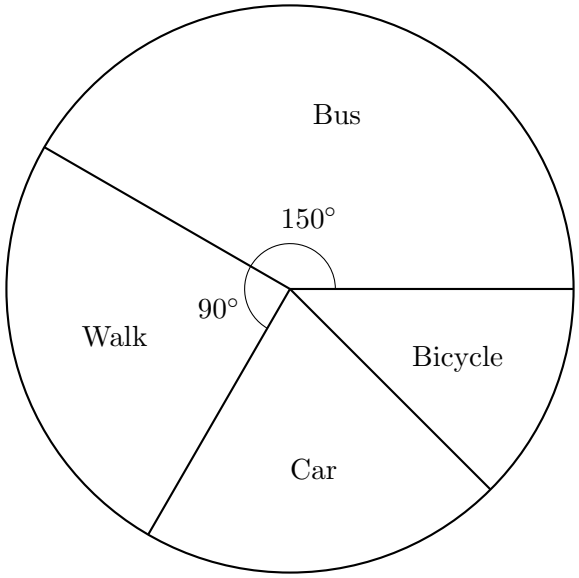
**(b)** Find the median mass.

..... kg **[1]**

**(c)** Work out the range.

..... kg **[1]**

**8** The pie chart shows information about how 120 students travel to school.




**(a)** Calculate the number of students who travel by Bus.

..... **[2]**

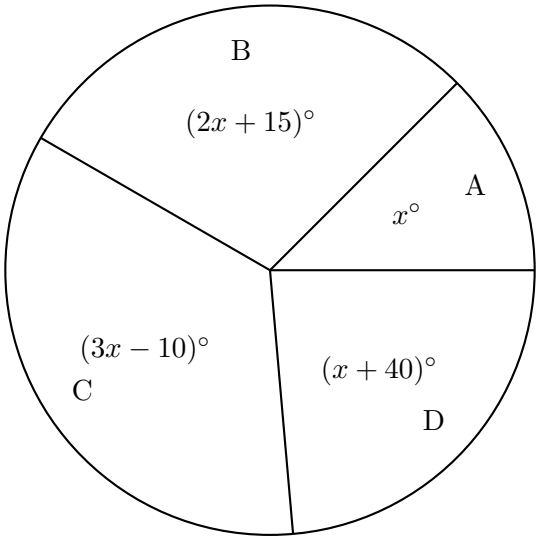
**(b)** There are 25 students who travel by Car.
Calculate the sector angle for Car.

..... ° **[2]**

**(c)** Calculate the percentage of students who travel by Bicycle.

..... % **[3]**

**9** A pie chart shows the grades (A, B, C, and D) achieved by students in a mathematics test.
The angles of the sectors are given in terms of $x$.




**(a)** Show that $x = 45$.

**[3]**

**(b)** There are 24 students who achieved Grade A.
Calculate the total number of students who took the test.

..... **[2]**

**(c)** Calculate the percentage of students who achieved Grade C.

..... % **[3]**

**1** **(a)** 21 **[B1]**

**(b)** 25 **[B1]**

**(c)** 33 **[B1]**

**(d)** 6 (parcels identified) **[M1]**

$\frac{6}{15}$ or $\frac{2}{5}$ **[A1]**

**[Back to Question]** *[Total Marks: 5]*

**2** **(a)** $360 - (110 + 80 + 70 + 40) = 60$ **[B1]**

**(b)** $\frac{80}{360} \times 180$ **[M1]**

40 **[A1]**

**(c)** $\frac{110}{360}$ **[M1]**

$\frac{11}{36}$ **[A1]**

**[Back to Question]** *[Total Marks: 5]*

**3** **(a)** Correct stems $1, 2, 3, 4$ **[B1]**

Correct ordered leaves for at least two stems **[B1]**

Fully correct ordered diagram **[B1]**

**(b)** 2.9 **[B1]**

**(c)** $\frac{4}{15}$ **[B1]**

**[Back to Question]** *[Total Marks: 5]*

**4** **(a)** $\frac{35}{120} \times 360$ **[M1]**

105 **[A1]**

**(b)** Sector angle for Tennis $= \frac{25}{120} \times 360 = 75°$ **[B1]**

$\frac{75}{360} \times \pi \times 6^2$ **[M1]**

23.6 (accept 23.56 to 23.57) **[A1]**

**(c)** $\frac{54}{108} \times 360$ or $54 \div \frac{108}{360}$ **[M1]**

180 **[A1]**

**[Back to Question]** *[Total Marks: 7]*

**5** **(a)** $2x + 30 + x + x + 50 + 2x - 20 = 360$ **[M1]**

$6x + 60 = 360$ leading to $x = 50$ **[A1]**

**(b)** Angle for Tennis $= 100°$ **[M1]**

$\text{Total} = 10 \times \frac{360}{100} = 36$ **[A1]**

**(c)** Number of students choosing Football $= 13$ and Basketball $= 5$ **[B1]**

$\frac{13}{36} \times \frac{5}{35} \times 2$ **[M1]**

$\frac{13}{126}$ **[A1]**

[Back to Question] *[Total Marks: 7]*

**6** **(a)** Total Laptops $= 50 + 60 + 70 = 180$ **[B1]**

Total Tablets (Jan + Feb) $= 40 + 60 = 100$ **[B1]**

$180 = 1.2 \times (100 + x)$ or $\frac{180}{1.2} = 100 + x$ **[M1]**

$x = 50$ **[A1]**

**(b)** Total items (Jan to Mar) $= 180 + 150 = 330$ **[B1]**

$\text{Total items (April)} = 330 \times \frac{1}{3} = 110$ **[M1]**

$\text{Laptops (April)} = \frac{6}{6+5} \times 110$ **[M1]**

60 **[A1]**

**(c)** Tablet revenue $= 40 \times 200 = 8000$ and Laptop revenue $= 50 \times 450 = 22500$ **[M1]**

$\frac{8000}{8000 + 22500} \times 100$ **[M1]**

26.2 (accept 26.22 to 26.23) **[A1]**

[Back to Question] *[Total Marks: 11]*

**7** **(a)** Correct stems used **[B1]**

Unordered leaves all correct **[B1]**

Fully correct ordered leaves:
Row 1: 2 5
Row 2: 3 7 7
Row 3: 1 4 8
Row 4: 0 2 2 5 9
Row 5: 1 3 **[B1]**

**(b)** 34 **[B1]**

**(c)** 41 **[B1]**

[Back to Question] *[Total Marks: 5]*

**8** **(a)** $\frac{150}{360} \times 120$ **[M1]**

50 **[A1]**

**(b)** $\frac{25}{120} \times 360$ **[M1]**

75 **[A1]**

**(c)** Angle for Bicycle $= 360 - 150 - 90 - 75 = 45°$ (or finding 15 students) **[M1]**

$\frac{45}{360} \times 100$ or $\frac{15}{120} \times 100$ **[M1]**

12.5 **[A1]**

**[Back to Question]** *[Total Marks: 7]*

**9** **(a)** $x + 2x + 15 + 3x - 10 + x + 40 = 360$ **[M1]**

$7x + 45 = 360 \implies 7x = 315$ **[M1]**

$x = 45$ **[A1]**

**(b)** $\frac{45}{360} \times \text{Total} = 24$ or $24 \times \frac{360}{45}$ **[M1]**

192 **[A1]**

**(c)** Angle for Grade C $= 3(45) - 10 = 125°$ **[M1]**

Percentage $= \frac{125}{360} \times 100$ **[M1]**

34.7 (accept 34.72 or 34.722...) **[A1]**

**[Back to Question]** *[Total Marks: 8]*